Advanced Internet of Things Technology

Week 4: Connectivity I – Matter & Thread, Messaging protocols in IoT

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Lecture positioning and objectives

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Positioning in the course

- Week 2: Device-level intelligence (TinyML)
- Week 3: Collaborative intelligence and orchestration
- Week 4: Connectivity as the enabler of intelligence

Lecture theme

- Solving the “Tower of Babel” in local IoT networks
- Why connectivity is the foundation of intelligent IoT systems
- Focus on interoperability, reliability, and local-first design

Learning objectives

- Understand why early IoT connectivity fragmented
- Explain how Matter and Thread address interoperability
- Compare MQTT and CoAP for constrained devices
- Make informed connectivity design decisions

The interoperability crisis in early IoT

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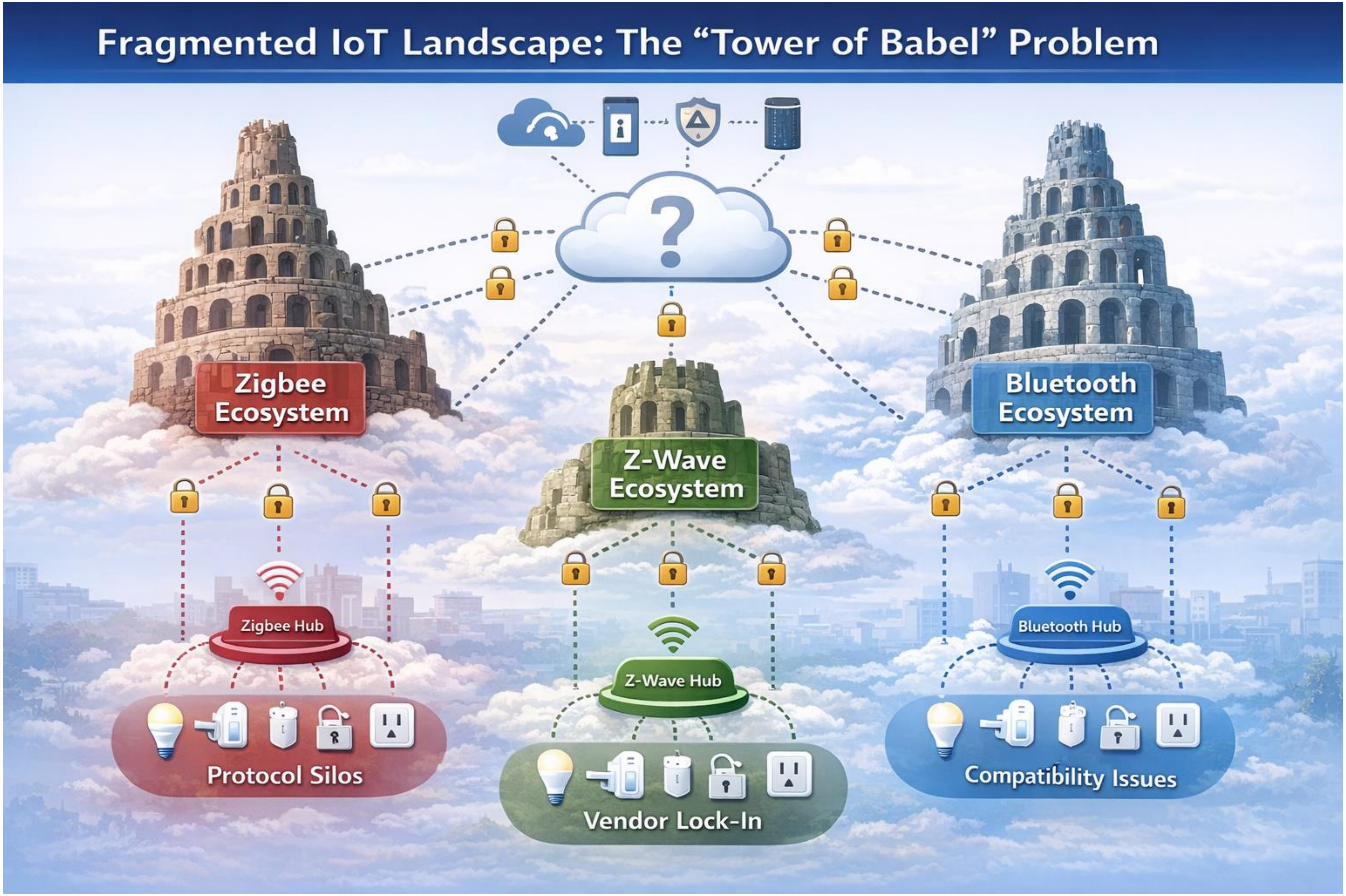
Fragmented IoT ecosystems

- Zigbee, Z-Wave, Bluetooth, Wi-Fi evolved independently
- Devices formed isolated protocol islands
- Lack of a common application-layer language

Consequences of fragmentation

- Poor user experience
- Limited cross-vendor automation
- Increased system complexity and cost


Fragmented IoT Landscape: The “Tower of Babel” Problem
Zigbee Ecosystem
Z-Wave Ecosystem
Bluetooth Ecosystem
Zigbee Hub
Z-Wave Hub
Bluetooth Hub
Protocol Silos
Vendor Lock-In
Compatibility Issues

Why Zigbee and Z-Wave failed to unify IoT

Protocol-level incompatibility

- Different PHY/MAC layers and frequencies
- Incompatible network and application layers
- Separate vendor alliances

Architectural limitations

- Non-IP-native design
- Dependency on protocol-specific hubs
- Difficult cloud and Internet integration

Bluetooth and Wi-Fi limitations

Bluetooth

- Designed for short-range, point-to-point communication
- Mesh support arrived late
- Limited suitability for whole-home networking

Wi-Fi

- High bandwidth but high energy consumption
- Poor fit for battery-powered sensors
- Cloud-centric application models

Root causes of the “Tower of Babel”

Lack of IP unification

- Most early protocols were non-IP
- Required protocol translation at gateways
- Increased latency and complexity

Inconsistent security and provisioning

- Different onboarding mechanisms
- Incompatible trust models
- Difficult unified management

Part 1: Matter standard

A technical standard for smart home and IoT (Internet of Things) devices.

Introduction to the Matter standard

Origin and vision

Project CHIP under the Connectivity Standards Alliance

Industry-wide collaboration

Goal: one application layer for smart devices

Design principles

IP-first and local-first

Interoperability by design

Security as a baseline requirement

https://handbook.buildwithmatter.com/how-it-works/what-is-matter/

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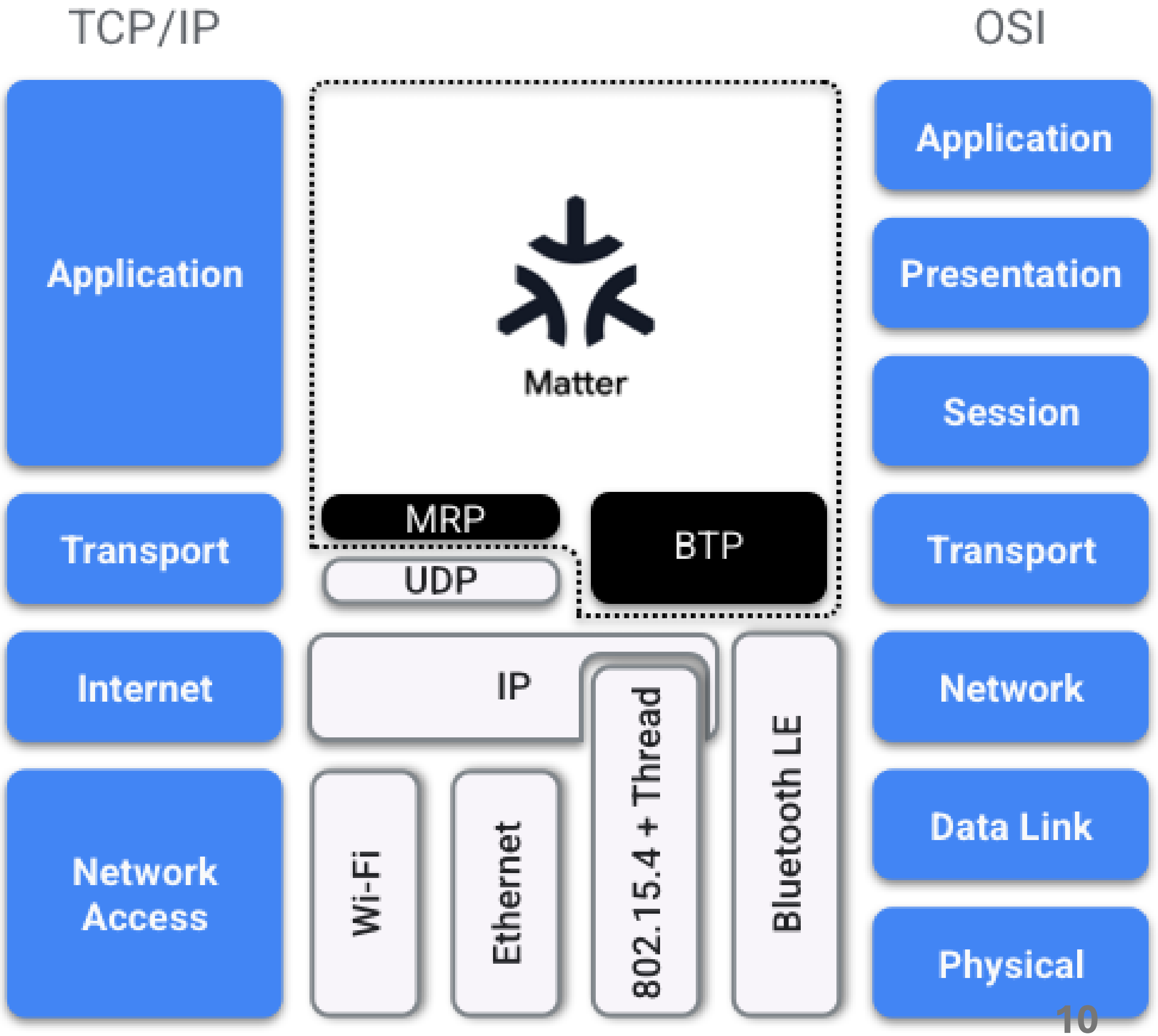
The Matter Network Stack

- Matter is flexible and interoperable.
- Matter also supports bridging of other Smart Home technologies such as Zigbee, Bluetooth Mesh and Z-Wave.
- Matter builds atop IPv6.
- It includes strong cryptography, a well-defined modeling of Device Types and their data, and the support for multiple ecosystem administrators.



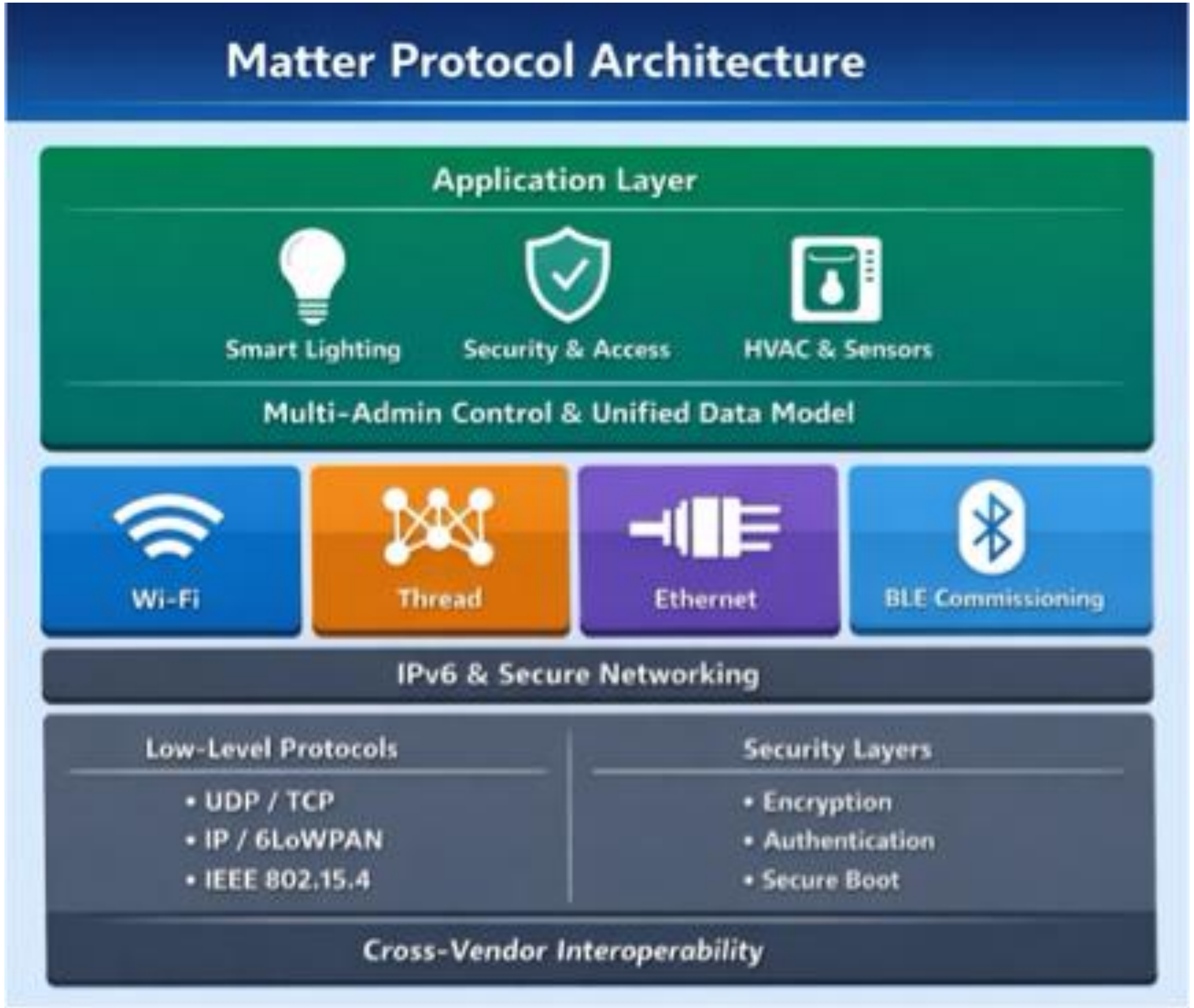
Matter architecture overview

Matter as a unified application layer

- Standardised device models and commands
- Runs over IPv6
- Independent of underlying transport

Supported transports

- Ethernet and Wi-Fi for high-bandwidth devices
- Thread for low-power mesh devices
- Bluetooth LE for commissioning



Matter Multi-Admin capability

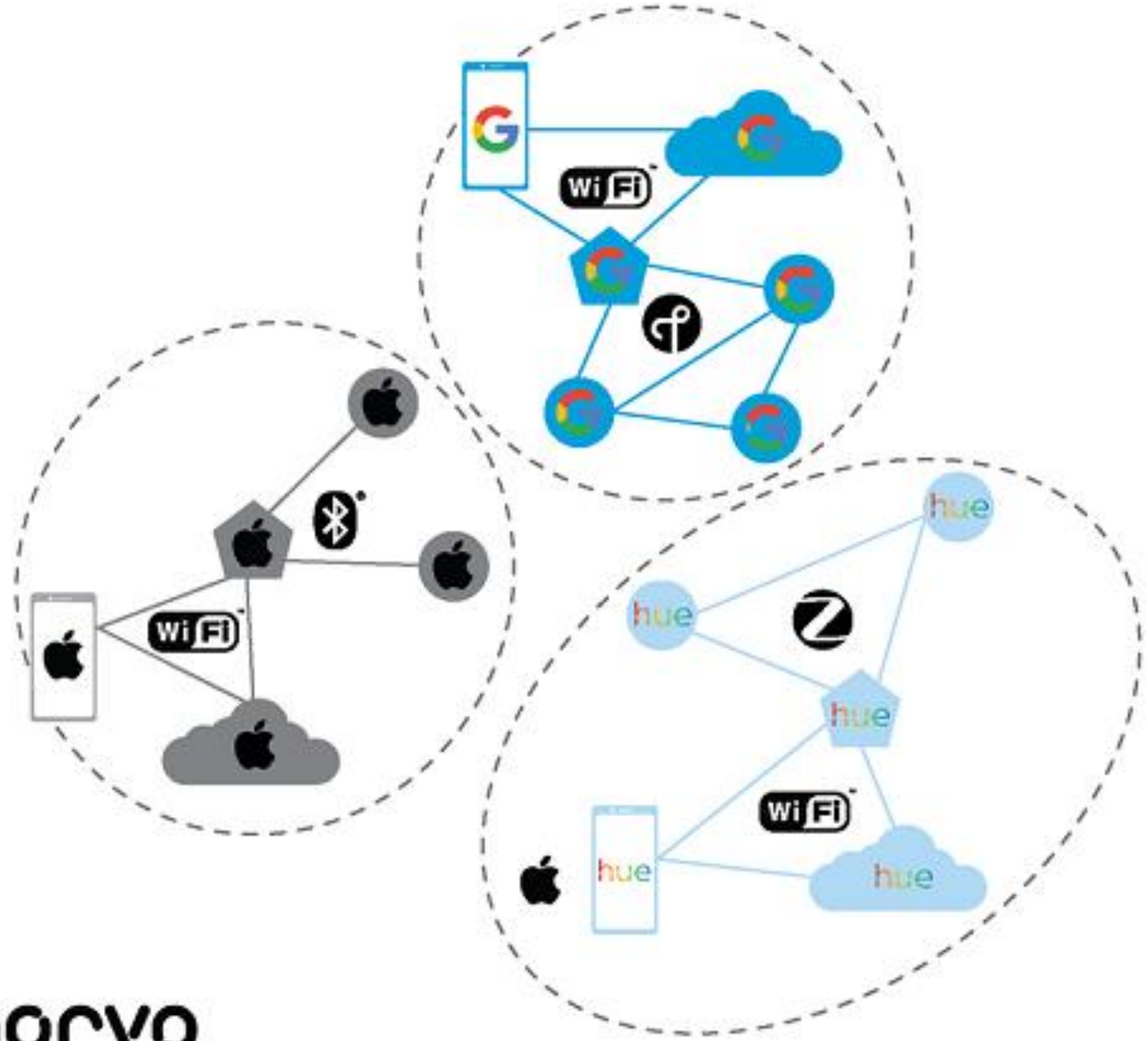
Motivation

- Users rely on multiple ecosystems
- Avoid vendor lock-in

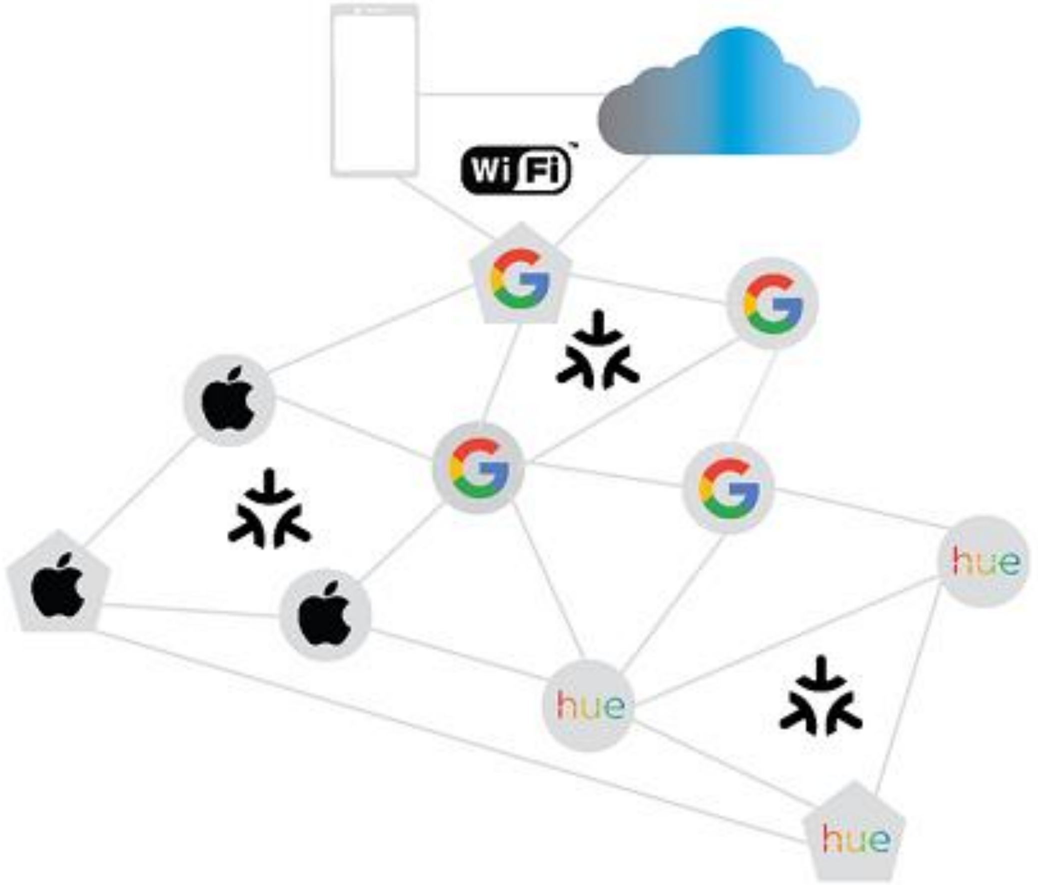
Multi-Admin concept

- Devices belong to multiple fabrics
- Multiple controllers operate simultaneously
- Local, secure coexistence

IoT Ecosystem: Today



IoT Ecosystem: Matter™



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Matter security and commissioning

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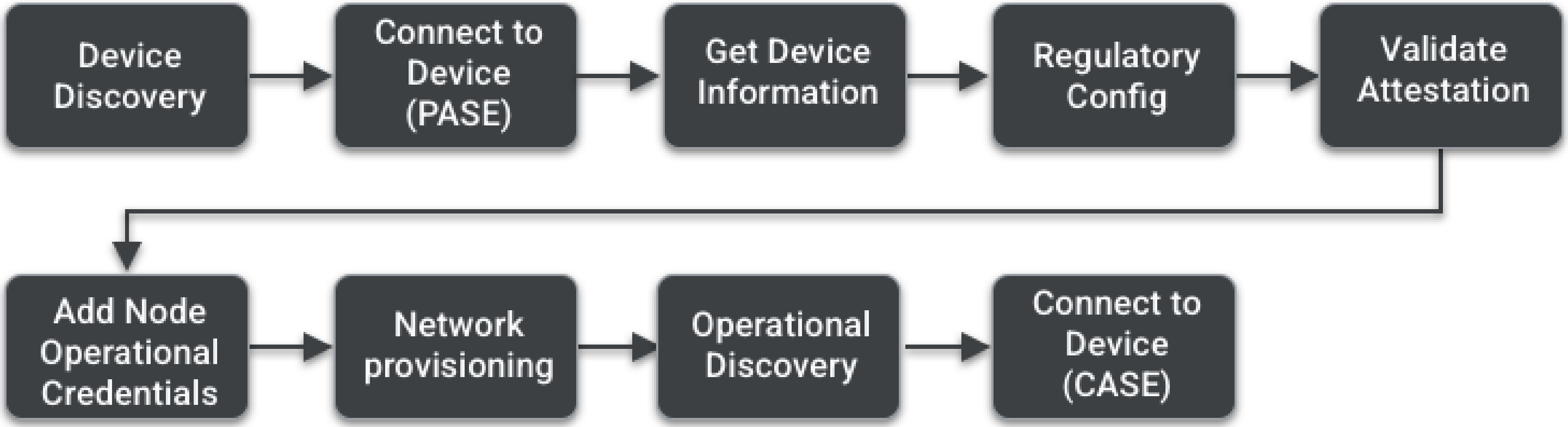
Secure onboarding

- Device attestation
- Encrypted commissioning process
- Trust established locally

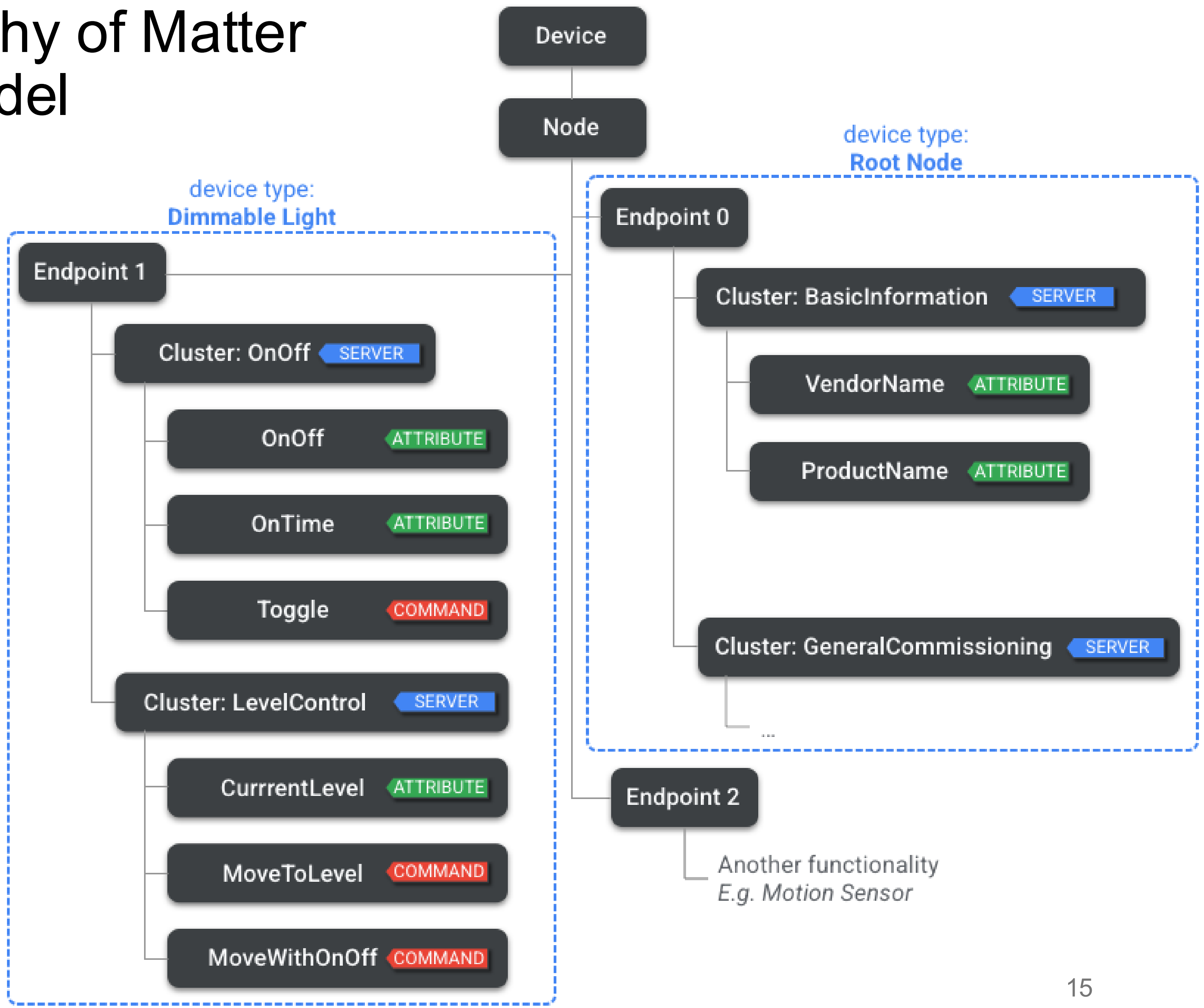
End-to-end security

- Encrypted device-to-device communication
- Consistent security across vendors

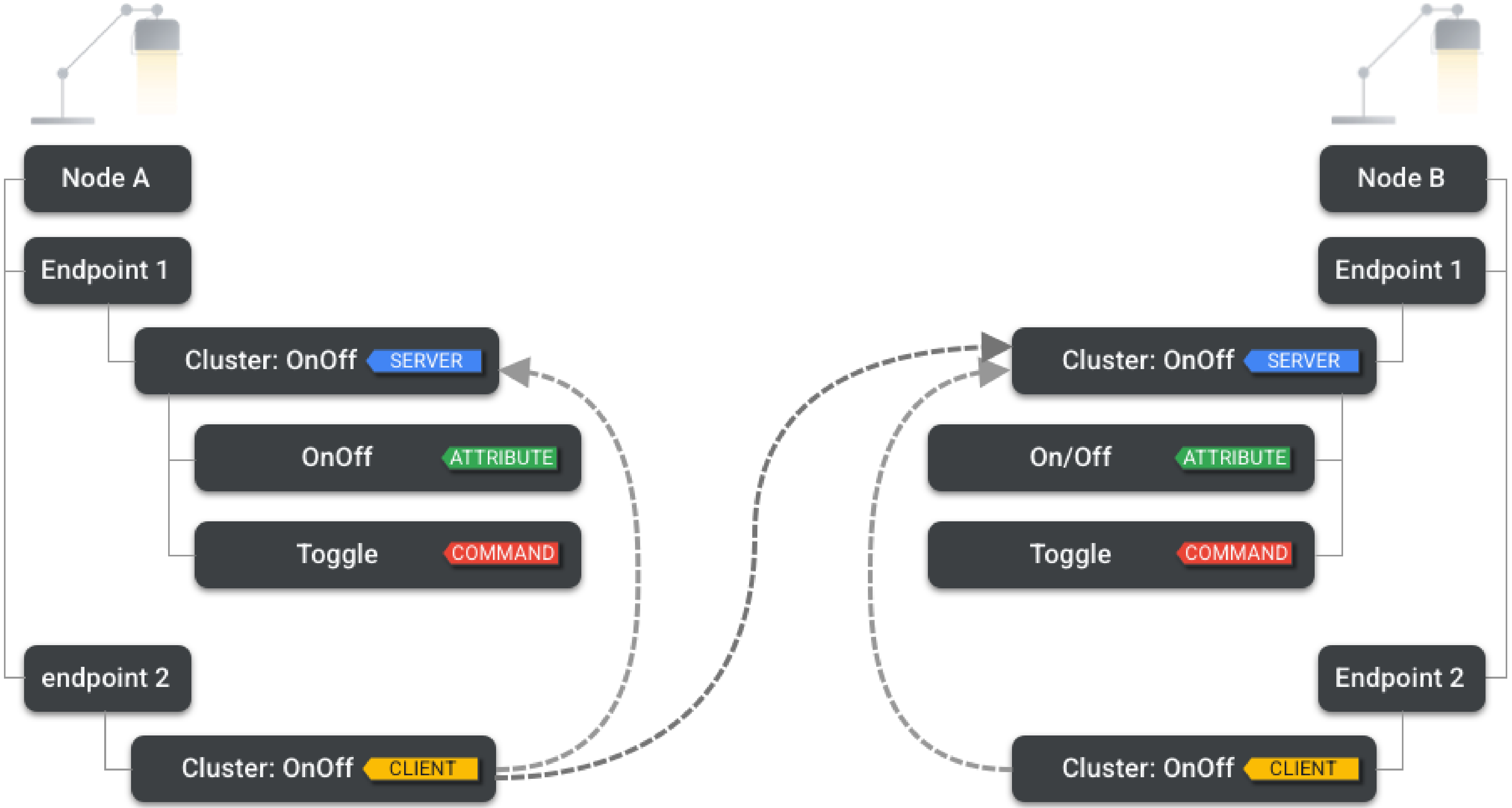


https://handbook.buildwithmatter.com/how-it-works/commisioning/

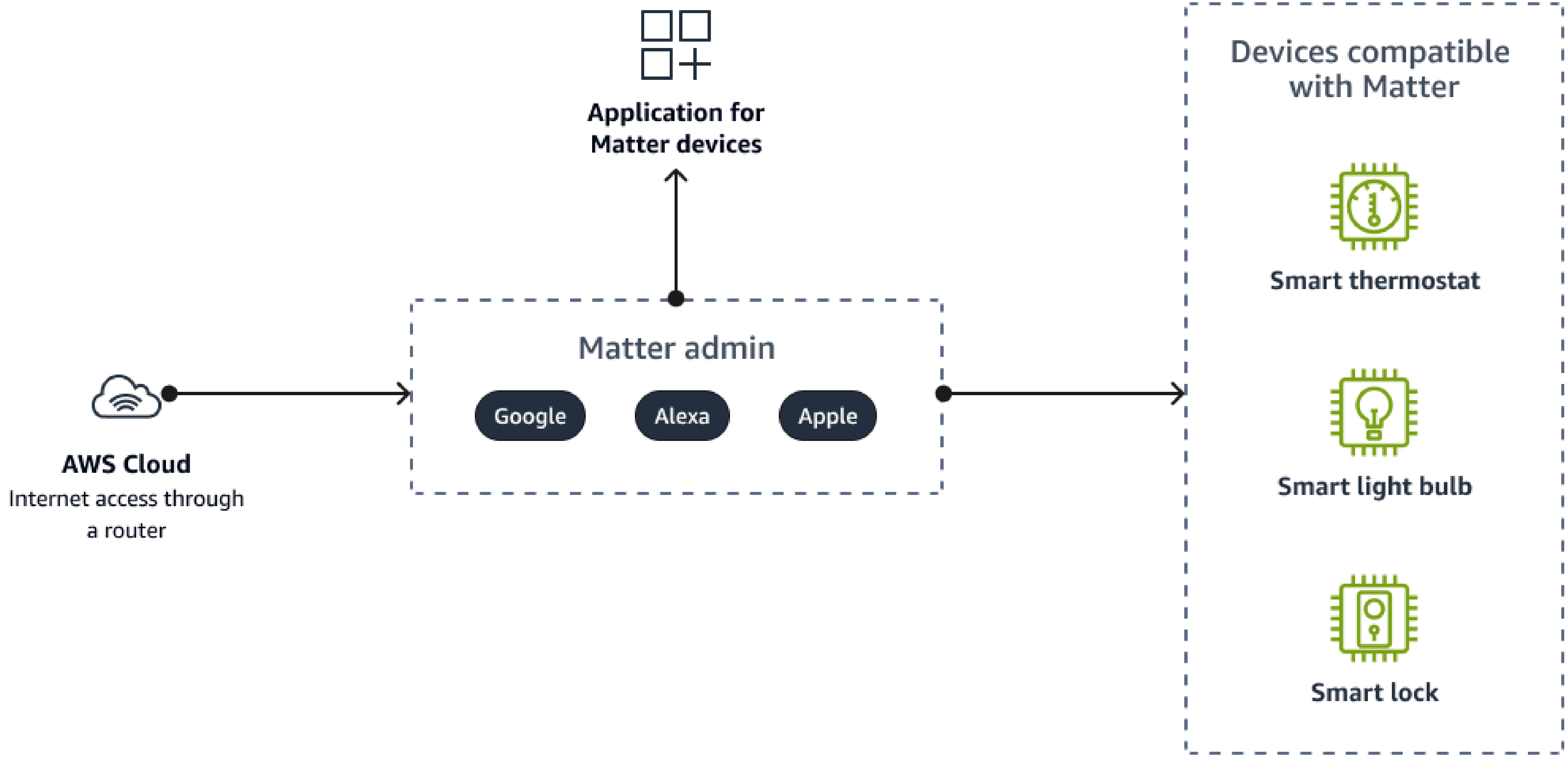
A sample of the hierarchy of Matter Devices interaction model



Client and Server Clusters



Use case


Application for
Matter devices
Matter admin
Google
Alexa
Apple
AWS Cloud
Internet access through
a router
Devices compatible
with Matter
Smart thermostat
Smart light bulb
Smart lock

Part 2: Thread protocol fundamentals

IPv6-based networking protocol designed for low-power Internet of Things

Thread protocol fundamentals

Why Thread matters

- Designed for low-power IoT
- Native IPv6 support
- Reliable mesh networking

Key characteristics

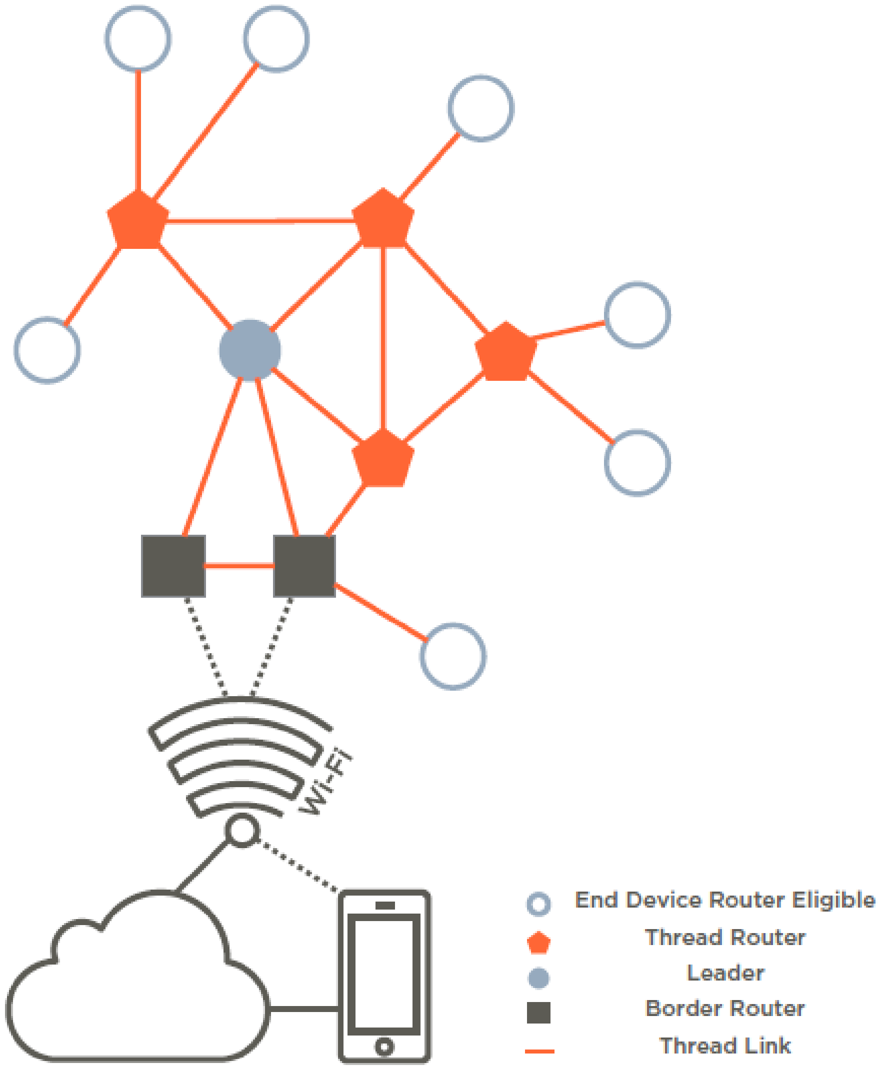
- Self-healing mesh
- Low latency and low power
- Scales to hundreds of devices

Thread General Characteristics

- Simple network installation, start-up, and operation
- Secure: Advanced Encryption Standard (AES) encryption
- Small and large home networks
- Bi-directional service discovery and connectivity.
- Typical devices provide sufficient range to cover a normal home.
- No single point of failure
- Low power
- Cost-effective

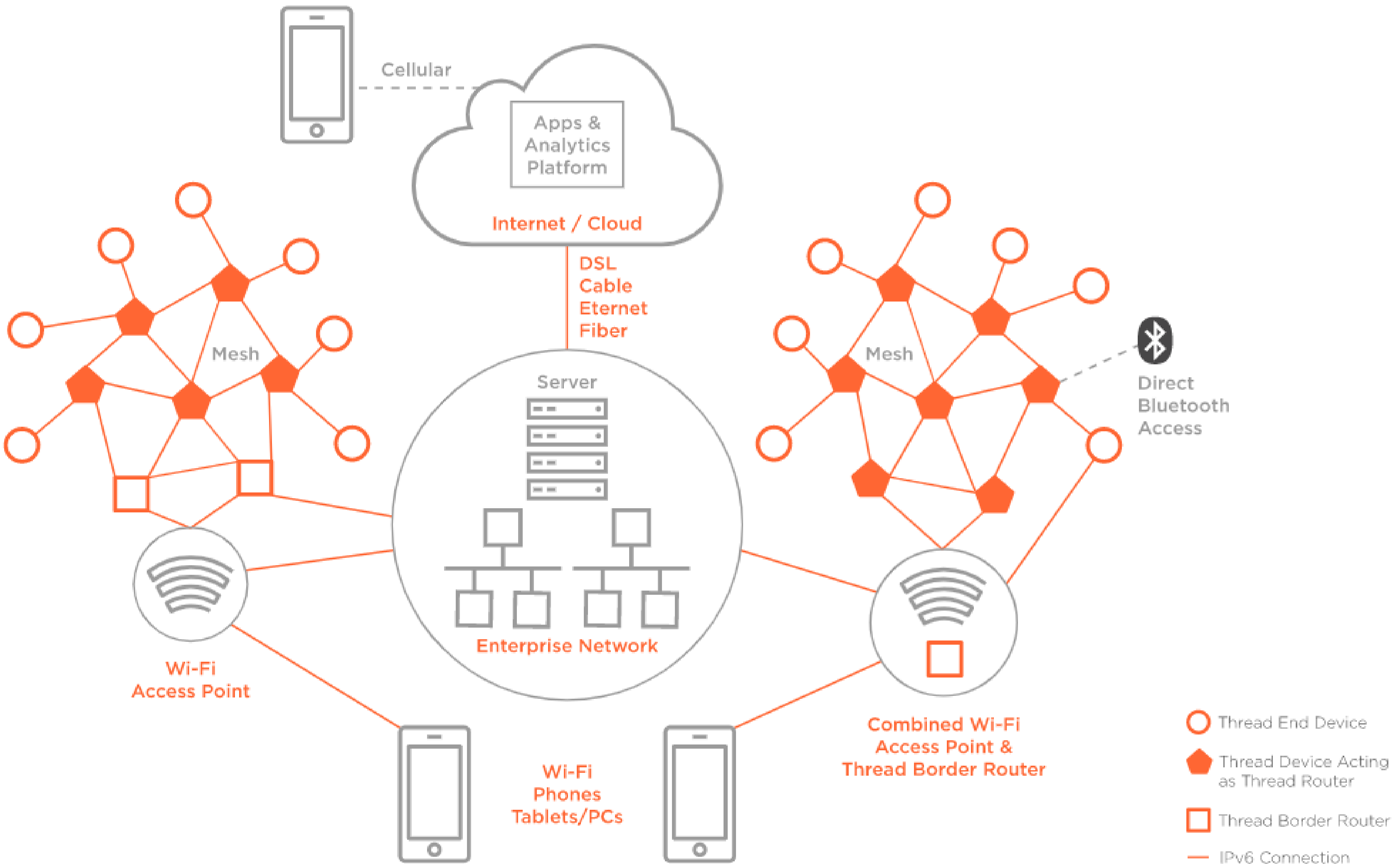
THREAD	Standard
Application Layer	
UDP + DTLS	RFC 768, RFC 6347, RFC 4279, RFC 4492v RFC 3315, 5007
Distance Vector Routing	RFC 1058, RFC 2080
6LowPAN (IPv6)	RFC 4944, RFC 4862, RFC 6775
IEEE 802.15.4 MAC (including MAC security)	IEEE 802.15.4 (2006)
IEEE 802.15.4 PHY	IEEE 802.15.4 (2006)

Thread mesh networking



- Mesh operation
  - Multi-hop routing
  - Dynamic role assignment
  - Automatic route repair
- Node roles
  - Leader, Router, End Device
  - Roles adapt dynamically

Commercial Network Topology

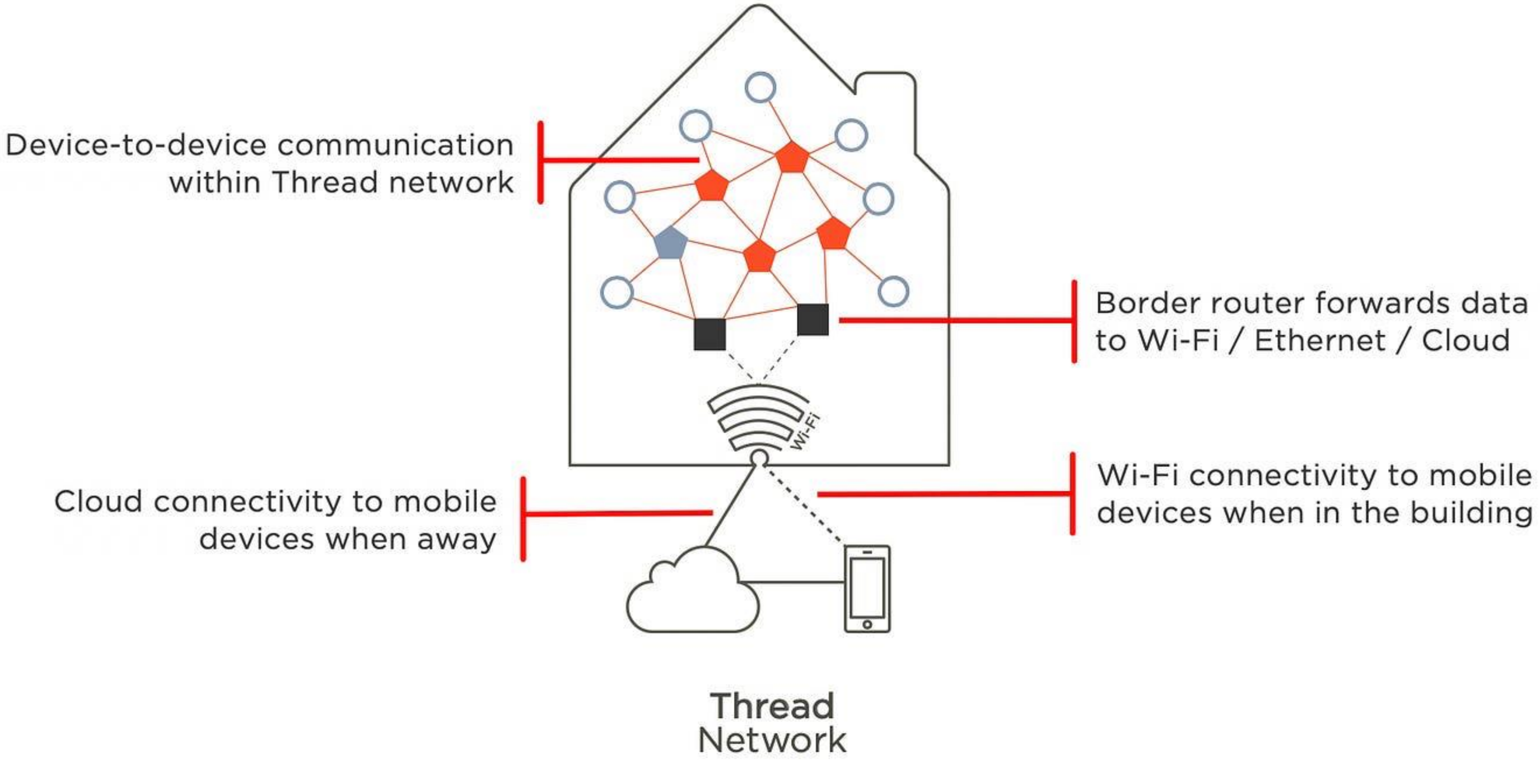


Thread Border Routers

- **Purpose**
  - Connect Thread mesh to IP networks
  - Enable Internet and Wi-Fi communication

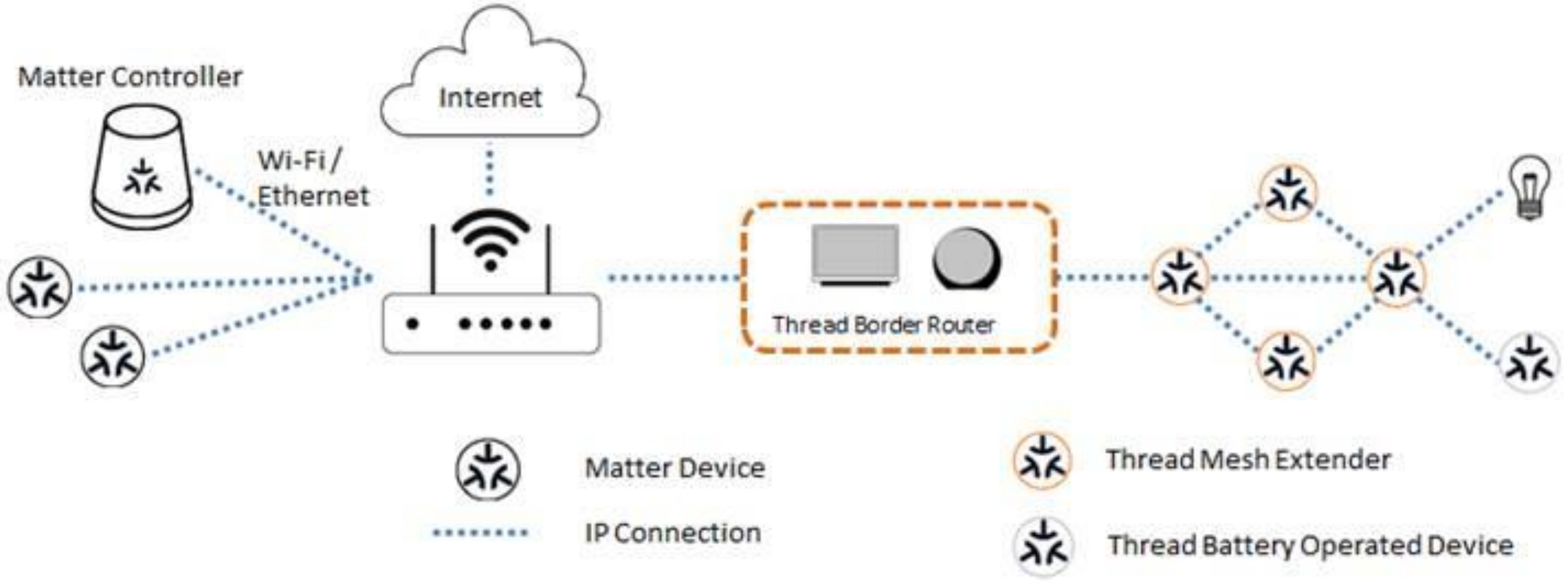
Key properties

- Multiple border routers supported
- No single point of failure
- Seamless failover

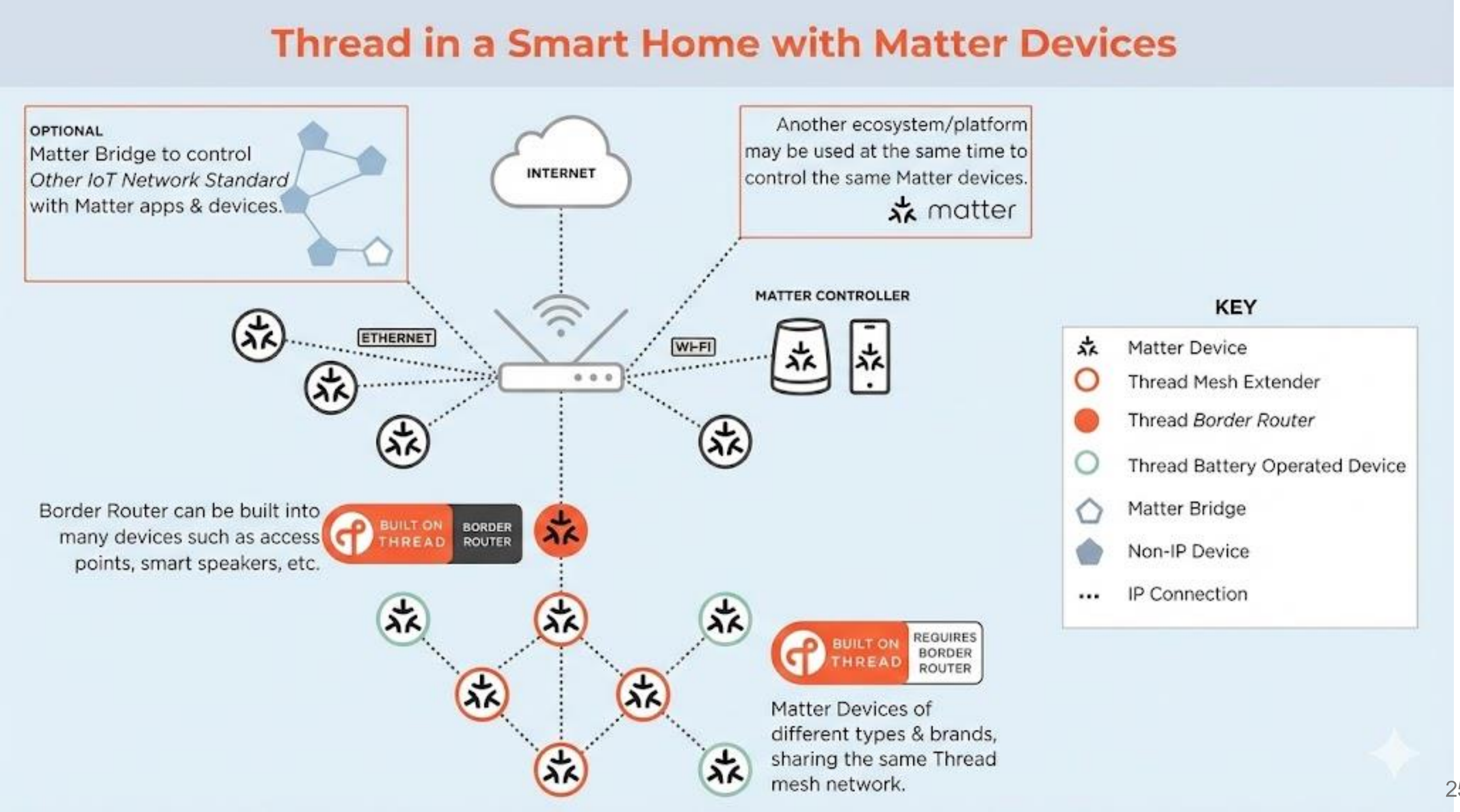


Matter + Thread end-to-end stack

- Separation of concerns
  - Thread handles networking
  - Matter handles application logic
- Benefits
  - Local control without cloud dependency
  - High reliability and responsiveness
  - Improved privacy



Case study


Thread in a Smart Home with Matter Devices
OPTIONAL
Matter Bridge to control
Other IoT Network Standard
with Matter apps & devices.
INTERNET
Another ecosystem/platform
may be used at the same time to
control the same Matter devices.
matter
MATTER CONTROLLER
ETHERNET
WI-FI
Border Router can be built into
many devices such as access
points, smart speakers, etc.
BUILT ON THREAD
BORDER ROUTER
BUILT ON THREAD
REQUIRES BORDER ROUTER
Matter Devices of
different types & brands,
sharing the same Thread
mesh network.
KEY
Matter Device
Thread Mesh Extender
Thread *Border Router*
Thread Battery Operated Device
Matter Bridge
Non-IP Device
IP Connection

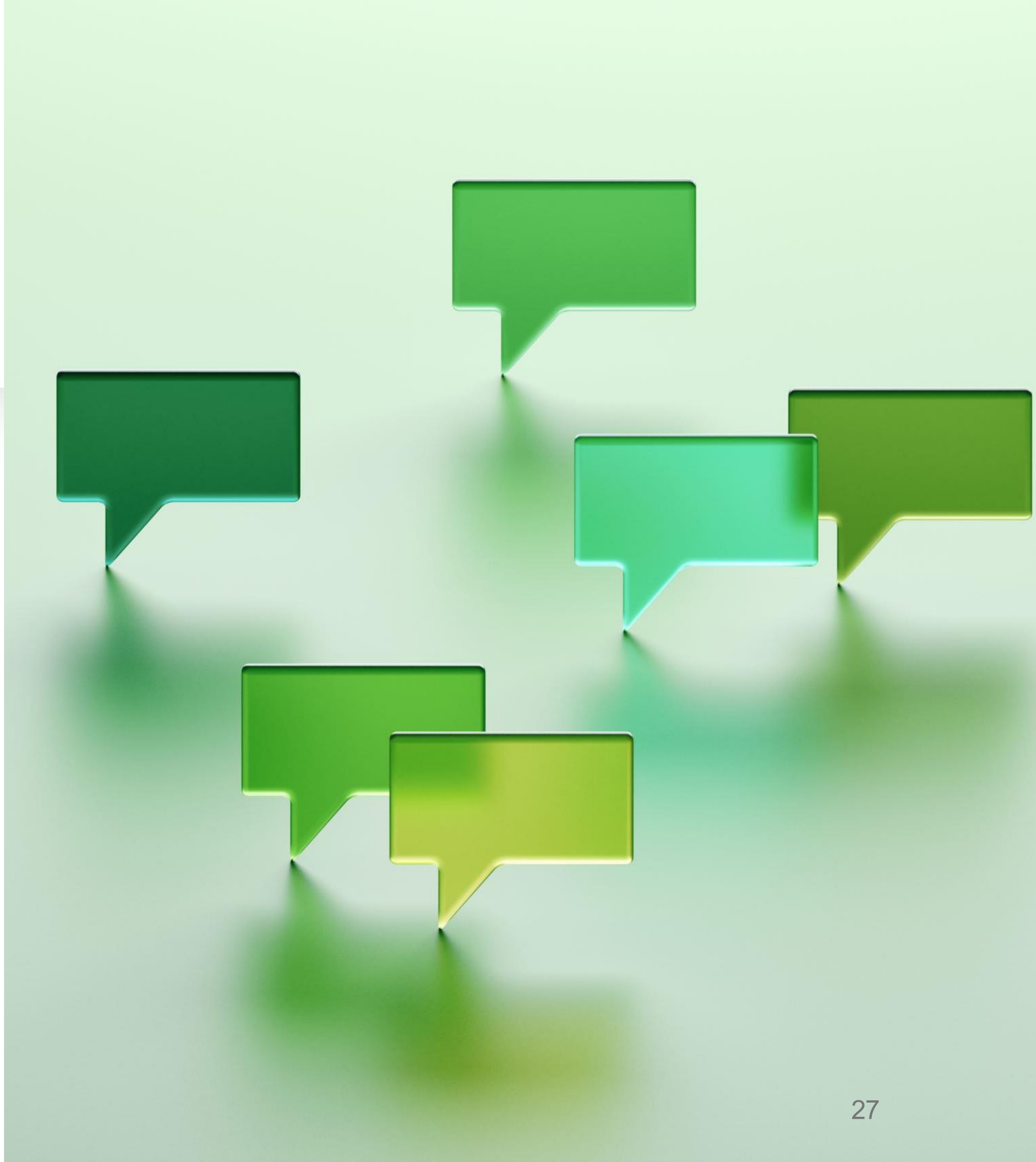
Part 3: Messaging protocols in IoT

Learn what are Messaging Protocols in IoT

Messaging protocols in IoT

Why messaging protocols matter

- Application-layer communication
- Device-to-device and device-to-cloud interactions
- Complement Matter-based systems



MQTT overview

Communication model

- Publish/subscribe via broker
- Decoupled producers and consumers

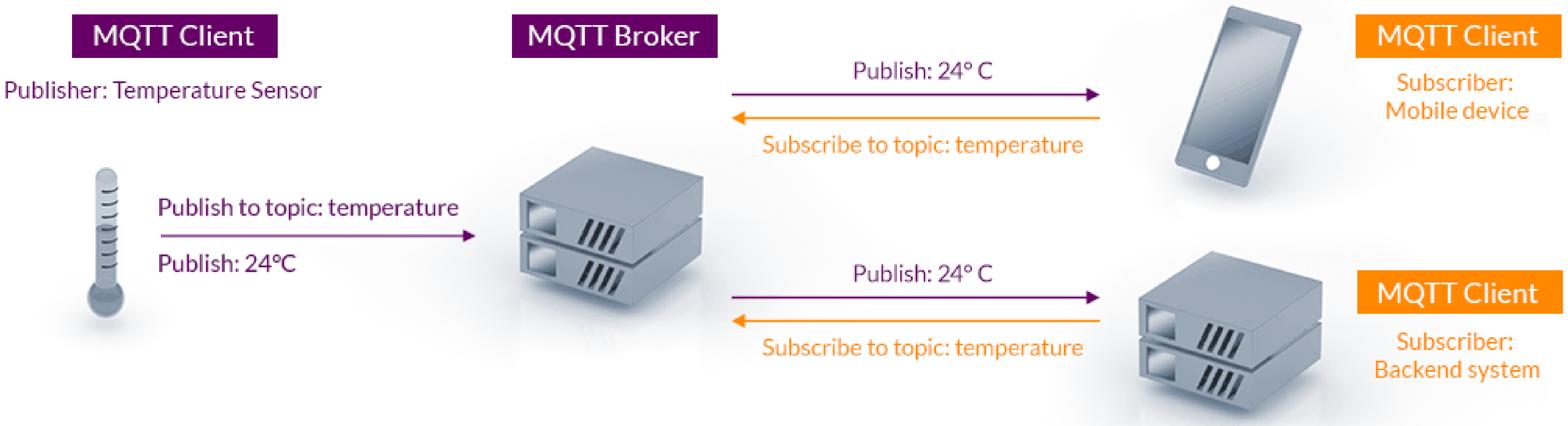
Strengths

- Scalable data distribution
- Cloud-friendly
- QoS-based reliability

Limitations

- Broker dependency
- TCP overhead for constrained devices

MQTT Publish/Subscribe Architecture



CoAP overview

Communication model

- Request/response (REST-like)
- Direct device interaction

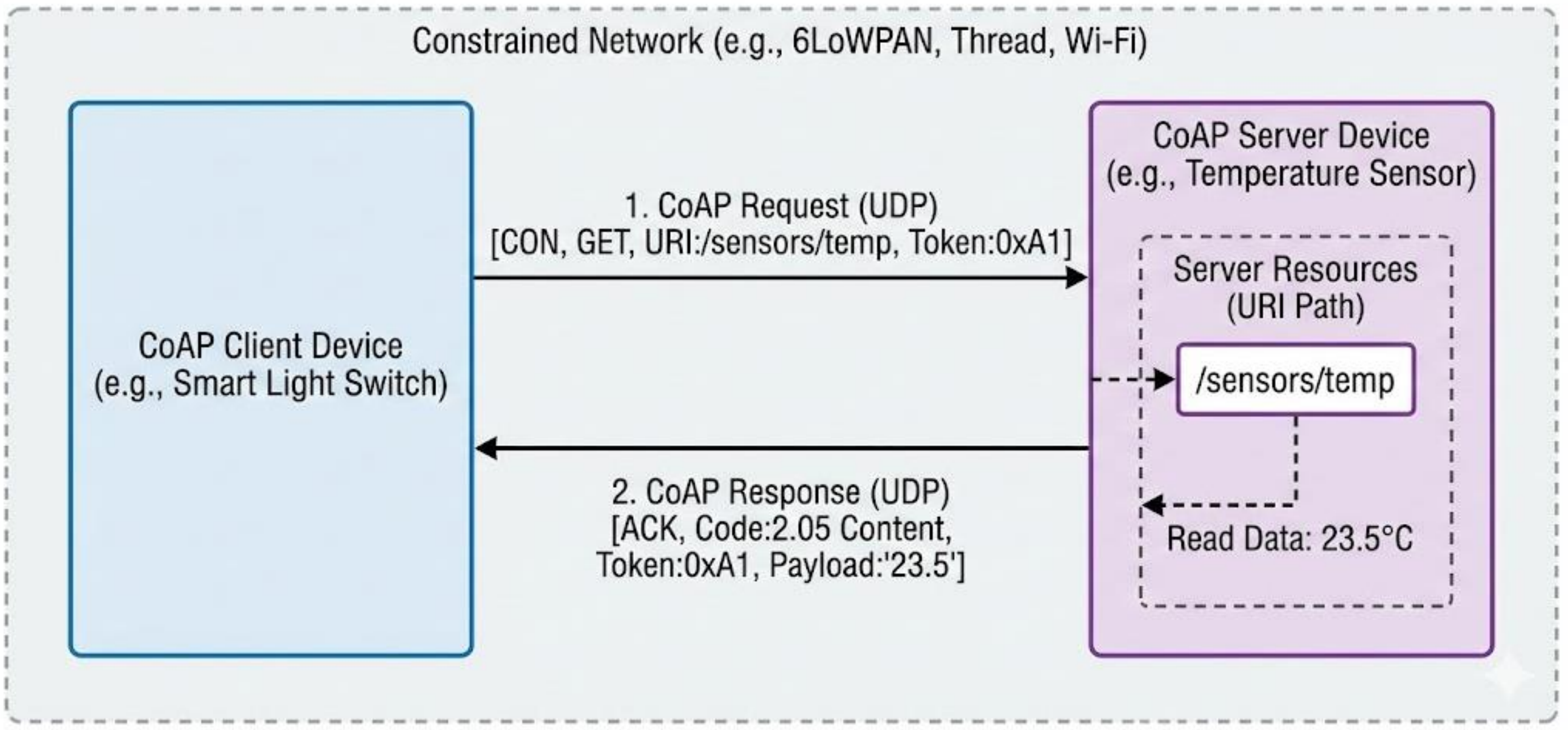
Strengths

- Lightweight UDP-based design
- Low energy consumption
- Suitable for constrained networks

Limitations

- Less scalable for many-to-many communication

CoAP Request/Response

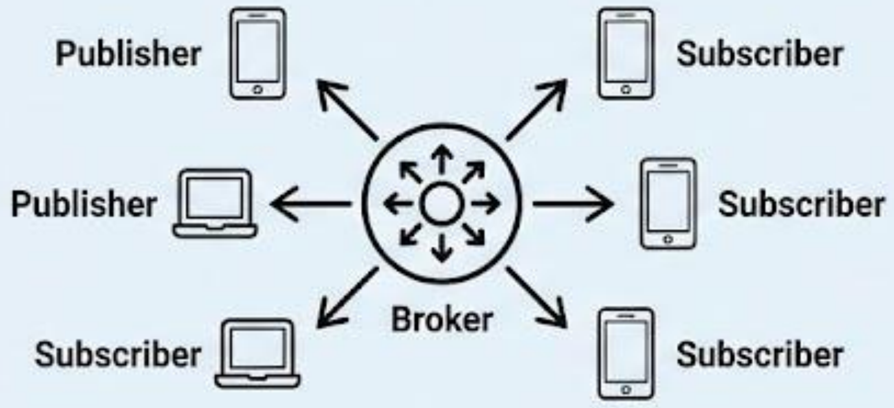


MQTT vs CoAP comparison

	MQTT	CoAP
Transport Layer	TCP	UDP
Header Size	2 bytes	4 bytes
Resource Overhead	Low	Very Low
Message Model	Pub/Sub	Request/Response RESTful
Message Reliability	Very High	Lower
Feature variety	Great	Lesser

MQTT vs. CoAP: IoT Protocol Comparison

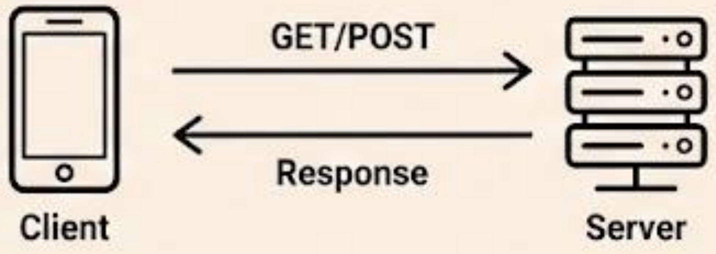
MQTT (Message Queuing Telemetry Transport)



TCP connection

- **Architecture:** Publish/Subscribe (Broker-based)
- **Transport:** TCP (Reliable, Connection-oriented)
- **Data Model:** Topics (Hierarchical Strings)
- **Overhead:** Higher (TCP handshake)
- **Best For:** Reliable cloud communication, complex routing.

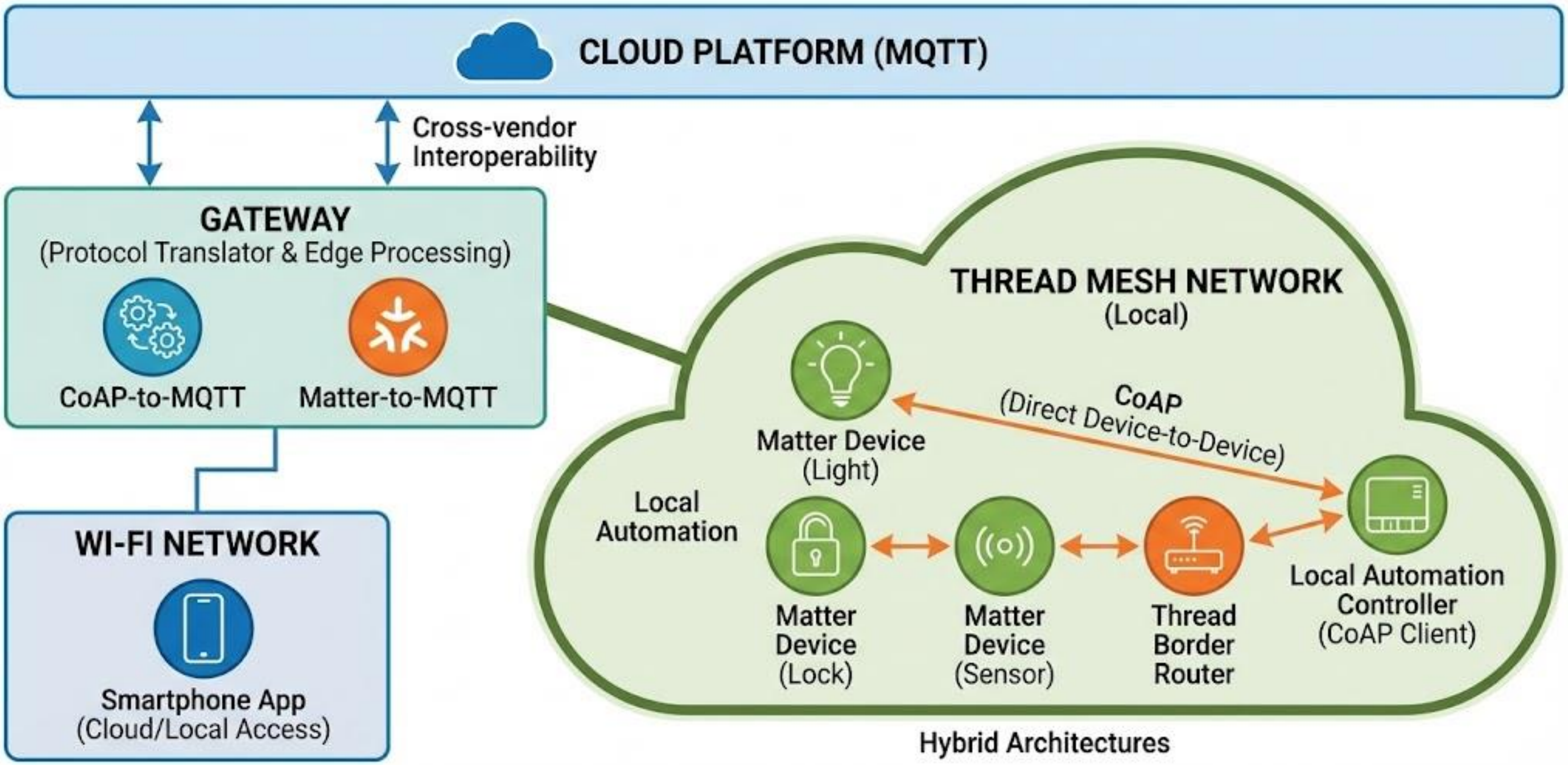
CoAP (Constrained Application Protocol)



UDP datagram

- **Architecture:** Request/Response (RESTful, P2P)
- **Transport:** UDP (Unreliable, Connectionless)
- **Data Model:** Resources (URIs)
- **Overhead:** Lower (4-byte header)
- **Best For:** Low-power, constrained networks (6LoWPAN), device-to-device.

Hybrid Smart Home IoT Architecture Case Study



Key design takeaways

Lessons for engineers

- Connectivity shapes system architecture
- Interoperability must be designed upfront
- Local-first communication improves reliability

Preparation for future systems

- Foundation for scalable edge intelligence
- Ready for integration with 5G/6G IoT

Closing and outlook

Summary

- Matter solves interoperability at the application layer
- Thread enables reliable low-power mesh networking
- MQTT and CoAP serve complementary roles

Next lecture preview

- Connectivity II: LPWAN and wide-area IoT integration

References

[1] https://handbook.buildwithmatter.com/how-it-works/what-is-matter/

[2] https://www.silabs.com/documents/public/user-guides/ug103-11-fundamentals-thread.pdf

[3] https://mqtt.org/

[4] https://datatracker.ietf.org/doc/html/rfc7252.